
DISCELL

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Abstract

A cell’s measured transcriptome in targeted in situ spatial transcriptomics mixes three things: what the cell intrinsically is, what varies with its microenvironment, and transcripts that were assigned to it but belong to its neighbours. The second and third are confounded at first order, since both make a cell’s expression predictable from its neighbours’ types, so no model fit to a single section can tell them apart without further assumptions. We present DISCELL, a variational model that splits a cell’s expression into an intrinsic latent state $\mathbf{z}$, a latent response $\mathbf{w}$ associated with a deterministic description of its microenvironment, and a fixed leakage mixture over graph neighbours governed by a leak fraction κ . Type-conditional invariance keeps niche-predictable information out of $\mathbf{z}$ through an adversarial regulariser, with a closed-form Gaussian penalty as the cheaper first attempt, and is graded by an independently trained held-out probe rather than by the training objective. Because κ is not identifiable from a single section, we do not estimate it: every model is refitted on a grid of κ values, and the deliverable is which spatial effects survive the grid. **[TODO: empirical claims to be added once the evaluation exists]**

1 INTRODUCTION

Imaging-based spatial transcriptomics assays such as Xenium (Janesick et al., 2023) measure thousands of transcripts per cell in situ, at single-cell resolution, over sections containing hundreds of thousands of cells. The appeal is obvious: a cell’s expression can be read together with the identity and state of the cells around it,

which is the raw material for asking how a microenvironment shapes the cells within it. The difficulty is that a cell’s measured count vector is not a clean readout of that cell. It is at least three things superimposed. First, the cell’s *intrinsic* state: its type, its cell-cycle phase, its clone-associated expression, and the systemic signals that reach every cell of its type regardless of position. Second, its *response* to the local microenvironment: the programmes that differ with who its neighbours are and what the surrounding tissue architecture looks like. Third, a *measurement artefact* specific to in situ assays: transcripts that were detected in the vicinity of a segmentation boundary and assigned to the wrong cell. Segmentation of densely packed tissue is imperfect, and published estimates place the fraction of misassigned transcripts anywhere between one tenth and one half **[NEEDS A SOURCE]**.

The second and third components are confounded at first order. A cell whose counts are contaminated by its neighbours acquires their markers; a cell that responds to its neighbours shifts its own programmes in a way their types predict. Where the responding genes are also markers of the neighbouring types, as for many ligands and receptors, or where a programme is shared across a region, the two produce the same signature, and neighbour composition predicts both. That misassigned transcripts produce apparent neighbourhood effects, including spurious cell–cell interaction signals, has been shown directly on in situ data (Mitchel et al., 2026), and the contamination a cell receives grows with the local density of the donor type (Bilous et al., 2026). The two differ in form, one adds a neighbour’s own profile, the other shifts the cell’s own programmes, but from the cell-by-gene count matrix alone the split between them is fixed only by the parametric forms one is willing to assume, and a flexible decoder can mimic either. Any claim that a spatial effect on expression is real therefore depends, silently, on how much of the neighbour-predictable signal was attributed to contamination.

This paper makes that dependence explicit rather than hiding it. We propose DISCELL (*disentangled cell states*), a generative model of per-cell counts with two

latent variables and one fixed mixing operator. A latent $\mathbf{z}_i$ carries the cell’s intrinsic state and is, by architecture and by regularisation, kept free of information about the microenvironment given the cell’s type. A latent $\mathbf{w}_i$ carries the variation associated with a deterministic context descriptor $\mathbf{c}_i$ that summarises who and what is nearby. A sparse, precomputed leakage kernel β mixes a fraction κ of each neighbour’s clean expression profile into the cell’s expected composition. The leak fraction κ is not estimated. It is fixed, the model is refitted on a grid of κ values, and every spatial effect is reported as a function of κ with its envelope over random seeds, a sensitivity analysis in which κ is the sensitivity parameter. An effect that is stable across the grid and across seeds is a finding; one that changes sign or vanishes is not separable from leakage on this data. The range across the grid is the deliverable, reported alongside an operating point. It is not a confidence interval, and it does not shrink with more cells.

Our modelling choices follow from three requirements that we motivate in Section 2 and Appendix B. The decoder receives no type label, so that $\mathbf{z}$ is forced to carry identity and cannot degenerate into a residual code. The context encoder queries the neighbourhood with the cell’s type rather than with $\mathbf{z}_i$ and reads only the neighbours’ types, closing a channel through which attention would otherwise reconstruct the cell from its most similar neighbours and make the prior on $\mathbf{w}$ ego-informed. And the invariance we enforce is *conditional* on type: cell types cluster in space, so a marginal invariance between $\mathbf{z}$ and neighbourhood composition deletes cell type from $\mathbf{z}$ altogether, the trade-off known from demographic parity (Zhao and Gordon, 2019) and observed for conditional subspace VAEs (Slavutsky et al., 2026).

Related work. Node-centric expression models (Fischer et al., 2023) regress a cell’s expression on its neighbourhood directly and produce a niche descriptor without a per-cell latent; our context descriptor $\mathbf{c}_i$ plays that role, and $\mathbf{w}_i$ is what remains after conditioning on it. Several models share our design goal of separating intrinsic from spatially associated variation. SIMVI (Dong et al., 2025) splits a cell’s state into an intrinsic and a spatially induced part with an independence regulariser; MintFlow (Akbarnejad et al., 2025) conditions an intrinsic latent on cell type and a microenvironment latent on neighbour-type composition, and discourages the former from predicting the latter with type-specific discriminators; Celcomen (Megias et al., 2026) separates intra- from inter-cellular gene-interaction programmes with a generative graph neural network; and NicheCompass (Birk et al., 2025) characterises niches with a graph neural network. None of them models transcript leakage, the confound this paper is about, and Mint-

Flow names its susceptibility to segmentation errors as a limitation [TODO: verify the NicheCompass characterisation; add scVIVA and further spatial VAEs]. Contamination-aware models for in situ data, notably resolVI (Ergen et al., 2025), model transcript misassignment explicitly and estimate a per-cell mixing weight; we instead treat the section-wide leak fraction as un-identified and sweep it. resolVI also disables gradients through the neighbour path because the coupling otherwise collapses the fit; we adopt the same stop-gradient and describe the degeneracy it prevents. Variational disentanglement of condition-specific from shared representations (Klys et al., 2018; Slavutsky et al., 2026) supplies the two-bound objective and the posterior-regularisation view of an adversarial invariance term (Ganchev et al., 2010) that we build on. We depart from them in three ways. The condition is not an observed label but the cell’s niche, summarised by a deterministic context descriptor on which the prior of $\mathbf{w}$ is conditioned. The invariance is conditional on cell type rather than marginal, because type and niche are dependent in tissue, as is standard in fair and domain-invariant representation learning (Madras et al., 2018; Li et al., 2018; Pogodin et al., 2023). And our intrinsic path keeps the full decoder and draws $\mathbf{w}$ from its prior instead of decoding from $\mathbf{z}$ alone (Appendix A.2). Our invariance target is a fixed data descriptor of the niche, so the model cannot shape it to make invariance trivial, unlike an independence penalty between two learned latents (Dong et al., 2025). That adversarial removal is not sufficient without an independent check is the observation of Elazar and Goldberg (2018); our probe protocol is its application. [TODO: extend related work: segmentation and contamination correction methods, further spatial VAEs and GNN spatial models]

Contributions. (i) A generative model that decomposes in situ counts into intrinsic state, microenvironmental response and neighbour leakage, with a multinomial likelihood in which the leakage mixture is exact under Poisson sources whose leaked-in rate scales with the receiving cell’s own rate rather than the donors’ (Section 2.2). (ii) An inference scheme whose architecture closes information channels that a regulariser cannot reach on its own: the intrinsic encoder never sees the context, the attention context is queried by type, and neighbour quantities are frozen, with an exact two-hop batching scheme that makes the frozen path affordable (Sections 2.3 and 2.6). (iii) A conditional-invariance regulariser, adversarial in its operating configuration with a closed-form penalty as the cheaper first attempt, conditional on type as in the type-specific discriminators of Akbarnejad et al. (2025) but targeting an image-derived niche descriptor along-

side neighbour composition, together with a held-out probe protocol that grades either independently of the training objective (Section 2.5). (iv) The leakage sweep as a sensitivity analysis: spatial effects are reported as trajectories over κ with seed envelopes, alongside an operating point (Section 2.8). **[TODO: (v) the empirical evaluation, to be added]**

2 THE DISCELL MODEL

We present the model in the order in which its parts depend on one another: the observed quantities and the spatial graph (Section 2.1); the generative model and the assumptions behind it (Section 2.2); the inference networks (Section 2.3); the objective (Section 2.4); the conditional-invariance regulariser and the held-out probe that grades it (Section 2.5); the frozen-neighbour batching that makes the model trainable (Section 2.6); model selection (Section 2.7); and what is and is not identified, which is what turns a fit into a finding through the leakage sweep (Section 2.8). Each design decision is stated with its reason; the longer arguments are in Appendix B and the derivations in Appendix A.

2.1 Setting and Notation

We consider a single tissue section profiled with a targeted in situ assay. For each of N segmented cells i we observe a count vector $\mathbf{x}_i \in \mathbb{N}^G$ over G genes, its total $\ell_i = \sum_g x_{ig}$, a cell-type label $t_i \in \{1, \dots, K\}$, and the morphology image around the cell. The label may be a curated annotation or a frozen unsupervised clustering and is treated as data, with two caveats that recur below: it was itself derived from the same contaminated counts, and its granularity is a modelling choice that sets what the decomposition is relative to (Section 2.5). Curated annotations frequently encode state or location as well as lineage (tumour- versus stroma-associated fibroblasts, hypoxic or proliferating tumour subsets). Such labels are niche-informed: conditioning on them assigns the corresponding response to t , where neither $\mathbf{w}$ nor the invariance of Section 2.5 sees it. Labels should therefore be lineage-level, and state-bearing subtypes merged before fitting **[TODO: the current fits use state-bearing curated labels or fine unsupervised clusters; merge to lineage level before the final sweeps]**. The total ℓ_i is conditioned on throughout and never modelled.

Spatial graph. Cells are nodes; two cells are adjacent if and only if their Voronoi cells share a face, i.e. no third cell lies between them (Delaunay adjacency). This choice has no neighbourhood count k and no radius, adapts to local packing with an average degree near six, and uses only cell centroids, so it is

insensitive to segmentation-boundary noise and to the nuclear-expansion segmentation that in situ platforms fall back on. Delaunay adjacency also connects cells across lumens, vessels and tears, so edges longer than a cut-off ($40 \mu\text{m}$ here) are pruned. We write $\mathcal{N}_i$ for the pruned neighbour set of i , which excludes i itself; all its members are first-order neighbours, so no cell-size correction of distances is needed. The same graph serves every role below.

For each edge we record the centroid distance d_{ij} and the length f_{ij} of the shared Voronoi face, measured after clipping each Voronoi cell to a $30 \mu\text{m}$ disc around its centroid so that faces at tissue borders and across gaps are bounded. The face length is a proxy for contact extent that is defined whether or not the segmented polygons touch. We deliberately do not use the length of shared segmented boundary: segmented polygons of adjacent cells rarely touch, so a contact length would vanish on a large share of edges, and where it is non-zero it tracks the gap between polygons, which is local density, a property of the niche that we want to keep out of the *technical* part of the model, the part that describes measurement rather than biology (Appendix B.2, Table 2).

Derived observables. From the graph and labels we form the neighbour-type composition $\mathbf{y}_i \in \Delta^{K-1}$, $y_{ik} = |\{j \in \mathcal{N}_i : t_j = k\}|/|\mathcal{N}_i|$, which excludes the cell itself. From the image we form an embedding $\Phi_i \in \mathbb{R}^{d_\Phi}$ of a patch centred on the cell, produced by a frozen pretrained vision model, in which a disc covering the cell has been masked out, so that Φ_i describes the surrounding tissue and not the cell. The two descriptors act at different scales: the graph is one hop, a few to thirty micrometres, while the masked patch describes tissue beyond roughly $25 \mu\text{m}$; both are what we mean by “the niche”. Here d_Φ is the embedding’s native width; whether a low-dimensional projection should replace it is an open implementation choice (Appendix C.2). Three lookup tables are fixed before training: $\bar{\mathbf{y}}(t)$, the mean composition of cells of type t ; $\bar{\Phi}(t)$, the mean image-niche membership of type t ; and $\bar{\mathbf{v}}(t)$, the mean of the invariance block of Section 2.5 for type t . Isolated cells, those with no neighbours after pruning, have an empty composition and are excluded from all three tables.

Leakage kernel. The leakage kernel is a sparse matrix over the same edges, normalised over the neighbours of each connected cell,

$$\beta_{ij} \propto f_{ij} \exp(-d_{ij}/\tau), \quad \sum_{j \in \mathcal{N}_i} \beta_{ij} = 1, \quad (1)$$

with contamination decay length $\tau = 20 \mu\text{m}$; it is row-stochastic on connected cells, and an isolated cell keeps

an all-zero row. It is precomputed once and never learned. Continuous geometry is allowed here, and only here: β is not part of any conditioning variable and does not touch the estimand (Appendix B.2).

2.2 Generative Model

Each cell has two latent variables. The intrinsic state $\mathbf{z}_i \in \mathbb{R}^{d_z}$ carries what the cell is; the response $\mathbf{w}_i \in \mathbb{R}^{d_w}$ carries how cells of that type shift with their surroundings. The environment itself enters as a deterministic context descriptor $\mathbf{c}_i$ (defined in Section 2.3) that is shared by co-located cells. With m_ψ a learned prior-mean network, a a learned decoder, $\mathbf{B} \in \mathbb{R}^{G \times d_w}$ a learned loading matrix and $\kappa \in [0, 1)$ the fixed leak fraction,

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2)$$

$$\mathbf{w}_i \mid \mathbf{c}_i, t_i \sim \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (3)$$

$$\boldsymbol{\rho}_i = \text{softmax}(\mathbf{a}(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i), \quad (4)$$

$$\bar{\boldsymbol{\rho}}_i = \sum_{j \in \mathcal{N}_i} \beta_{ij} \boldsymbol{\rho}_j, \quad (5)$$

$$\mathbf{p}_i = (1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i, \quad (6)$$

$$\mathbf{x}_i \mid \ell_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i). \quad (7)$$

Here $\boldsymbol{\rho}_i \in \Delta^{G-1}$ is the cell's own *clean* composition, $\bar{\boldsymbol{\rho}}_i$ is the *foreign influx*, the leakage-weighted average of its neighbours' clean compositions, and $\mathbf{p}_i$ is the fitted multinomial probability. The decoder $\mathbf{a}(\mathbf{z}_i) \in \mathbb{R}^G$ is the cell's baseline programme in log space; column k of $\mathbf{B}$ is a gene programme and w_{ik} is how far cell i has moved along it. The prior variance is fixed at $\sigma_w = 1$, deliberately (Section 2.3).

For an isolated cell $\bar{\boldsymbol{\rho}}_i = \mathbf{0}$, and the mixture in Equation (6) would sum to $1 - \kappa$, charging the cell $-\ell_i \log(1 - \kappa)$ nats for leakage it cannot receive. The mixture is therefore renormalised per row,

$$\mathbf{p}_i = \frac{(1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i}{(1 - \kappa) + \kappa \sum_{j \in \mathcal{N}_i} \beta_{ij}}, \quad (8)$$

so that an isolated cell degenerates to $\mathbf{p}_i = \boldsymbol{\rho}_i$, i.e. $\kappa_i = 0$: no neighbours, no foreign source. For connected cells the denominator is one. Probabilities are floored at 10^{-8} inside the logarithm. Isolated cells keep an all-zero composition $\mathbf{y}_i$ and are not excluded from the regulariser or the probes of Section 2.5.

What the leakage term assumes. The leak fraction κ is section-wide, not per cell, and there is no ambient term. A cell's contamination in reality varies with its own size and depth, the depth and density of its neighbours, and how much boundary they share. The model scales the leaked rate with the receiving cell's own rate (below), so a low-count cell beside high-count donors is under-corrected at every κ , and a high-count

cell beside low-count donors over-corrected. The invariance of Section 2.5 keeps $\mathbf{z}$ free of anything niche-predictable, so the niche-predictable part of that error is absorbed by $\mathbf{w}$ and is not addressed by the sweep. Labels are derived from uncorrected counts and held fixed across the grid. Leakage-driven label errors concentrate where leakage is largest, at heterotypic interfaces and in low-count cells beside high-count neighbours; for such a cell the error enters through t_i and through its neighbours' compositions, where neither κ nor the invariance reaches it. Diffuse background is assumed negligible relative to leakage from adjacent cells [NEEDS A SOURCE]. The mixture in Equation (6) is exact under two assumptions: that the sources are independent Poisson variables, whose sum is again Poisson, and that the rate of leaked-in transcripts scales with the receiving cell's own rate rather than with the donors', which is where the per-cell variation just mentioned is hidden. The multinomial then follows from conditioning on the observed total (below). The parametrisation also assumes that leakage is gene-agnostic: κ carries no gene index, so a leaked transcript is drawn from the donor's whole-cell composition $\boldsymbol{\rho}_j$. Transcripts nearest the boundary, such as mRNAs of secreted and membrane proteins, are plausibly over-represented in what crosses it [NEEDS A SOURCE]. Such a gene-specific leak is not spanned by κ at any value and lands in $\mathbf{w}$, so a response programme enriched for cell-surface genes cannot be separated from leakage by the sweep alone.

What the decomposition means. The split between $\mathbf{z}$ and $\mathbf{w}$ is defined by two requirements. First, within a type, the intrinsic state should carry nothing that is predictable from the niche: $\mathbf{I}(\mathbf{z}_i; [\mathbf{y}_i, \Phi_i] \mid t_i) = 0$. In Equation (2) $\mathbf{z}_i$ is a priori independent of everything, and $\mathbf{y}_i, \Phi_i$ are not modelled, so this is not a property of the generative model; it is a constraint imposed on the variational family in Section 2.5, in the sense of posterior regularisation (Ganchev et al., 2010). Second, the environment reaches the counts only through the response and through leakage: $\mathbf{x}_i \perp \mathbf{c}_i \mid \mathbf{z}_i, \mathbf{w}_i, \{\boldsymbol{\rho}_j\}_{j \in \mathcal{N}_i}$, which holds by construction because the decoder of Equation (4) takes no context. Precisely, what the first requirement removes from $\mathbf{z}$ is variation predictable from the neighbour composition and the masked image given the type. It removes intrinsic variation that is spatially organised within a type only to the extent that neighbour composition and the masked image predict it. On a carcinoma section that extent can be large: subclones occupy contiguous regions and the masked image shows the nuclei of clonal kin, so part of a cell's clone-associated expression is predictable from its niche, is moved from $\mathbf{z}$ into $\mathbf{w}$, and would be read as a response; the part they do not predict stays in $\mathbf{z}$, as does variation carried by the neighbours' own

intrinsic codes. How much is at stake depends on the labels as well: a label that splits subclones keeps them in t , while under the lineage-level labels recommended above they are within-type variation, and on a carcinoma section this is then the main case rather than a corner case. This is why the decomposition is relative to t (Section 2.5).

Why a multinomial. If $x_{ig} \sim \text{Poisson}(\lambda_{ig})$ independently, then $\mathbf{x}_i \mid \sum_g x_{ig} = \ell_i$ is $\text{Multinomial}(\ell_i, \boldsymbol{\lambda}_i / \sum_g \lambda_{ig})$. Since ℓ_i is observed, the multinomial is the coherent likelihood: no library-size latent and no double counting of depth. Poisson sources are closed under addition, which is what makes the leakage mixture exact; negative-binomial sources with gene-specific means are not in general, and a negative binomial on the mixed counts would let the dispersion absorb part of what the mixture should explain. At the depth of in situ data, tens to a few hundred counts over thousands of genes, gene-wise overdispersion cannot be estimated in any case, and a low-dimensional amortised $\mathbf{z}$ absorbs much of what a dispersion parameter captures in coarser models. Should posterior predictive checks show overdispersion at higher depth, the appropriate extension is a Dirichlet-multinomial with one concentration scalar, not G dispersions.

Why κ is fixed. Leakage and response differ in form: one adds a neighbour’s own profile through a fixed contact kernel, the other shifts the cell’s own programmes along learned directions. From the cell-by-gene count matrix alone, however, the split between them is pinned only by those parametric forms, and a decoder flexible enough to fit the data can mimic either, since neighbour composition predicts both. We do not trust the likelihood to choose κ . Instead we fix it, fit the model independently at every point of a grid, and report each readout as a function of κ (Section 2.8). $\kappa = 0$ is the nested no-correction null. The grid is anchored at zero and extends to 0.4, close to the upper range of published misassignment estimates for this platform [NEEDS A SOURCE]; a data-driven upper bound from an atlas-defined marker set, and the transcript-level information that would identify κ from a nuclear/extranuclear split of the counts, are discussed in Appendix B.5.

Why the decoder has no type input. The decoder in Equation (4) takes $\mathbf{z}_i$ and $\mathbf{w}_i$ only, so the only route from identity to expression runs through $\mathbf{z}$. If $\mathbf{z}$ collapsed to a function of the label, both reconstruction terms in Section 2.4 would lose all within-type variance, a large fitting penalty, and that is what holds $\mathbf{z}$ open. With the label in the decoder, $\mathbf{z}$ would become a residual-only code and would no longer be interpretable as identity. What $\mathbf{z}$ carries beyond the label is

Table 1: Learned and fixed quantities. $d_z = 20$, $d_w = 6$; G , K and d_Φ are set by the panel, the label set and the image model.

Learned	Fixed before training
encoder of $q(\mathbf{z}_i \mid \mathbf{x}_i, t_i)$; attention layer and type query embed(t); prior mean m_ψ ; encoder of $q(\mathbf{w}_i \mid \cdot)$; decoder a ; loadings $\mathbf{B}$; adversary heads (separate optimiser)	graph, kernel β , τ ; leak fraction κ (swept); prior scale $\sigma_w = 1$; image embedding Φ and its principal components; image-niche memberships e^Φ ; tables $\bar{\mathbf{y}}(t)$, $\bar{\Phi}(t)$, $\bar{\mathbf{v}}(t)$; labels t ; totals ℓ

what the label discretises away: cell-cycle phase, clone-associated expression, position on a differentiation continuum, stress and metabolic state, states acquired in a niche the cell has since left, and systemic signals that affect every cell of a type regardless of position. The last is environmental in the colloquial sense but not niche-predictable, and it belongs in $\mathbf{z}$. Two remarks follow. Since the encoder of Section 2.3 receives the label, the agreement between $\mathbf{z}$ and t measures whether $\mathbf{z}$ retains type, not whether it discovers it; its collapse is the detector for the marginal-invariance failure mode (Section 2.7). And coarse labels leave more for $\mathbf{z}$ to carry; whether $\mathbf{z}$ separates cell-cycle phases within a type is an empirical check for the evaluation.

2.3 Inference Networks

Table 1 summarises what is learned and what is fixed. All networks are multilayer perceptrons unless stated; widths are in Table 3.

The context descriptor $\mathbf{c}_i$. One vector per cell summarising who and what is nearby. It is deterministic, is shared by co-located cells, and is the conditioning variable of the model. It is computed by a single graph-attention layer (Brody et al., 2022) over the pruned graph, concatenated with the image embedding and an isolation flag:

$$\mathbf{h}_j = \text{onehot}(t_j), \quad (9)$$

$$\mathbf{g}_i = \text{GATv2}(\text{query} = \text{embed}(t_i), \{\mathbf{h}_j\}_{j \in \mathcal{N}_i}), \quad (10)$$

$$\mathbf{c}_i = \mathbf{g}_i \oplus \Phi_i \oplus \mathbf{1}[\mathcal{N}_i = \emptyset]. \quad (11)$$

Here $\text{embed}(t_i)$ is a learned per-type vector of width K , the width of the source features; since the layer applies its own linear map to the query, this is equivalent to a one-hot query and is a parameterisation, not a design choice. The layer has no self-loops, no edge features, and its heads are averaged. Its attention weights π_{ij} satisfy $\sum_j \pi_{ij} = 1$. The context is therefore a pure function of data, types, graph and image; no latent enters it. With types as both query and sources, the logit of an edge depends on the two types alone, so the

attention output is a learned, type-specific reweighting of the neighbour composition, and $\mathbf{c}_i$ is a fixed function of $(t_i, \mathbf{y}_i, \Phi_i)$: the descriptors against which $\mathbf{z}$ is made invariant, so the prior on $\mathbf{w}$ sees exactly what $\mathbf{z}$ is kept from (Appendix B.2). Three decisions here are load-bearing.

The query is the type, not $\mathbf{z}_i$. In attention of this form the destination feature enters the attention logits but never the aggregated value, so using $\mathbf{z}_i$ as the query would inject no ego content directly. It would, however, open a selection channel: attention can learn $\pi_{ij} \propto \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j))$ and reconstruct the cell through its most similar neighbours, so that $\mathbf{c}_i$ becomes a mirror of $\mathbf{z}_i$. The incentive exists because an ego-informed prior on $\mathbf{w}_i$ is free decoder capacity, and it is strongest in homotypic regions, where κ is also least identified. The consequence would be that m_ψ already knows the cell, the divergence $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ collapses, and $\mathbf{w}_i$ measures deviation from *self* rather than from the expected response. Querying with the type closes the selection channel at no cost to the prior, since $m_\psi(\mathbf{c}_i, t_i)$ already receives t_i . The sources are the neighbours’ types only. An earlier version also fed the neighbours’ posterior means under a stop-gradient; removing them improved held-out reconstruction, type agreement and the mirror diagnostic in a three-seed ablation while leaving the response’s spatial readouts unchanged, so the input bought nothing measurable and was dropped. The within-type mirror coefficient is still monitored rather than assumed away (Section 2.7; Appendix B.1).

No per-edge geometry enters the context. The graph says who is adjacent; the image describes how packed and how structured the region is; per-edge distances and contact extents are withheld from $\mathbf{c}_i$. Three reasons, in decreasing order of weight. The concrete risk is collinearity: an attention layer given edge features can rediscover the leakage kernel β , after which $\mathbf{w}$ becomes collinear with $\bar{\rho}$ and the two mechanisms the sweep is meant to separate merge inside the model (Appendix B.2). Within a first-order neighbourhood the distances span perhaps 5 to 30 μm , most of which is local density, which the image already carries. And the intended reading of the context is that neighbour-type composition is the exposure of interest while the image and the graph are conditioning covariates; a hypothetical change in composition is a meaningful question about a tissue in a way that “the same neighbours, further away” is not, and putting per-edge geometry into the conditioning variable would mix the two. The estimand is used only informally here; counterfactual machinery is future work (Section 4).

Isolated cells. A cell with no in-edges has an empty attention set; the graph part of $\mathbf{c}_i$ is set to zero and

the flag is raised. The image part stays, since Φ_i is measured data defined for every cell. Together with Equation (8), an isolated cell is modelled with $\kappa_i = 0$ and an image-only context.

Posterior over $\mathbf{z}$. The posterior $q(\mathbf{z}_i | \mathbf{x}_i, t_i)$ is a diagonal Gaussian produced by a network whose input is the library-normalised, log-transformed count vector together with the log total and the one-hot type,

$$[\log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i), \log \ell_i, \text{onehot}(t_i)], \quad \tilde{\ell} = \text{median}_i \ell_i. \quad (12)$$

The encoder *never sees* $\mathbf{c}_i$: invariance by architecture, which the regulariser of Section 2.5 then completes. Library normalisation before the log matters because a raw $\log(1 + \mathbf{x})$ entangles depth with composition, so that a 300-count and a 50-count cell of the same type look very different; since ℓ_i is conditioned on rather than modelled, $\mathbf{z}$ should not spend capacity on it. Total counts are nonetheless weakly informative about state, so $\log \ell_i$ enters as one extra scalar rather than being discarded.

Prior and posterior over $\mathbf{w}$. The prior is what a typical cell of this type would do in this context; the posterior is what this cell did:

$$p(\mathbf{w}_i | \mathbf{c}_i, t_i) = \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (13)$$

$$q(\mathbf{w}_i | \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i) = \mathcal{N}(\boldsymbol{\mu}_{w,i}, \text{diag } \boldsymbol{\nu}_{w,i}^2), \quad (14)$$

with $\boldsymbol{\nu}_{w,i}$ the learned posterior scale, as distinct from the fixed prior scale σ_w . Both distributions are functions of the context, but only the posterior sees the cell itself, through the sampled $\mathbf{z}_i$ and through the encoded counts of Equation (12); we abbreviate the posterior as $q(\mathbf{w}_i)$ where its arguments are clear. The divergence $\text{KL}(q \| p)$ reads as how far this cell deviates from its expected response, an anomaly score that the objective provides at no extra cost; whether it carries usable per-cell information at the operating point is an empirical question for the evaluation. The prior is also what would make counterfactuals possible, by drawing $\mathbf{w}' \sim p(\mathbf{w} | \mathbf{c}', t)$ and decoding, with abduction $\mathbf{w}'_i = \mathbf{w}_i + m_\psi(\mathbf{c}', t_i) - m_\psi(\mathbf{c}_i, t_i)$ carrying the cell’s own residual unchanged: because the prior scale is fixed, the per-cell counterfactual is exactly a translation of the prior mean, and the difference $\mathbf{w}'_i - \mathbf{w}_i$ is manifestly free of the per-type gauge of Section 2.8. The type-level form of this machinery, in which the prior mean alone is moved to a new context and the cell’s intrinsic state is held fixed, is exercised in the evaluation; the per-cell form with abduction is not, since it presupposes a non-degenerate response posterior.

The direct path from $\mathbf{x}_i$ into the posterior is necessary. The conditional independence $\mathbf{w} \perp \mathbf{x} | \mathbf{z}, \mathbf{c}$ is false by explaining-away: $\mathbf{x}$ is generated from both $\mathbf{z}$ and $\mathbf{w}$,

so knowing $\mathbf{x}$ and $\mathbf{z}$ informs $\mathbf{w}$ directly, which is also why Slavutsky et al. (2026) condition their condition-specific posterior on the data rather than on the shared code. In our setting $\mathbf{z}$ is lossy in exactly the wrong direction: it is low-dimensional, the invariance regulariser removes niche-predictable variation from it, and that variation *is* the spatial response that $\mathbf{w}$ exists to capture. Without the direct path, $\mathbf{w}_i$ collapses to $\mathbb{E}[\text{response} \mid \text{type}, \text{niche}]$, identical for co-located similar cells, and the divergence vanishes (Appendix B.3).

Learned and fixed variances. Both posteriors carry learned diagonal variances; this is what makes the reparameterised sample meaningful (Kingma and Welling, 2014; Rezende et al., 2014) and lets uncertainty vary per cell, so that a 30-count cell can have a wider $q(\mathbf{z})$ than a 300-count cell. The prior scale σ_w is fixed at one, for two reasons of which the first is decisive. A learned *constant* σ_w is a gauge freedom: $\mathbf{w}$ and $\mathbf{B}$ enter the decoder only as $\mathbf{B}\mathbf{w}_i$, so rescaling dimension k of $\mathbf{w}$ by s and column k of $\mathbf{B}$ by $1/s$ leaves the likelihood unchanged, and a learned constant prior scale would absorb the compensating change in the prior; fixing it pins the scale. A learned *conditional* $\sigma_w(\mathbf{c}, t)$ would be meaningful, claiming that some niches produce more variable responses than others, but it opens a degenerate direction: the model can shrink $\text{KL}(q \parallel p)$ for free by inflating $\sigma_w(\mathbf{c}, t)$ wherever the penalty bites, making the prior vague rather than making $\mathbf{w}$ predictable. If it is wanted, it must be bounded (Appendix B.6).

Decoder. The decoder is Equation (4): a network $a(\mathbf{z}_i) \in \mathbb{R}^G$ plus the linear shift $\mathbf{B}\mathbf{w}_i$, followed by a softmax over genes. It takes no type input, for the reason given in Section 2.2. If reconstruction is poor the remedy is to widen $\mathbf{z}$ or the network, not to add t .

2.4 Objective

With the neighbour quantities $\mathbf{c}_i$ and $\bar{\rho}_i$ held constant (Section 2.6), the evidence lower bound on $\log p(\mathbf{x}_i \mid \mathbf{c}_i, t_i, \bar{\rho}_i)$, conditional on the neighbours’ clean compositions (the coupled model is treated in Appendix A.1), under the variational family $q(\mathbf{z}_i) q(\mathbf{w}_i \mid \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i)$ is

$$\begin{aligned} L_i = \mathbb{E}_{q(\mathbf{z}_i)} & \left[\mathbb{E}_{q(\mathbf{w}_i \mid \mathbf{z}_i)} \left[\log p(\mathbf{x}_i \mid \mathbf{z}_i, \mathbf{w}_i) \right] \right. \\ & - \text{KL} (q(\mathbf{w}_i \mid \mathbf{z}_i) \parallel p(\mathbf{w}_i \mid \mathbf{c}_i, t_i)) \Big] \\ & - \text{KL} (q(\mathbf{z}_i) \parallel p(\mathbf{z}_i)), \end{aligned} \quad (15)$$

where the $\mathbf{w}$ -divergence sits inside the outer expectation because $q(\mathbf{w}_i \mid \cdot)$ conditions on the sampled $\mathbf{z}_i$; evaluating the closed-form Gaussian divergence at the one reparameterised draw is an unbiased one-sample estimate of it (Appendix A.1). A second bound on the same evidence is obtained by restricting the variational

family to $q(\mathbf{w}_i) := p(\mathbf{w}_i \mid \mathbf{c}_i, t_i)$, which zeroes that divergence:

$$\begin{aligned} L_i^z = \mathbb{E}_{q(\mathbf{z}_i)} \mathbb{E}_{p(\mathbf{w}_i \mid \mathbf{c}_i, t_i)} & \left[\log p(\mathbf{x}_i \mid \mathbf{z}_i, \mathbf{w}_i) \right] \\ & - \text{KL} (q(\mathbf{z}_i) \parallel p(\mathbf{z}_i)). \end{aligned} \quad (16)$$

We call this the *intrinsic path*, and it is load-bearing. The first reconstruction term does reward $\mathbf{z}$ for carrying signal, since $a(\mathbf{z})$ is the only route to all G genes, but with the counts also wired into $q(\mathbf{w})$ the allocation between $\mathbf{z}$ and $\mathbf{w}$ is ambiguous, and the second bound breaks the tie toward $\mathbf{z}$: under it $\mathbf{z}$ must explain the cell given only the *expected* response. Because it runs through the same leakage mixture as the first bound, it never asserts $\kappa = 0$ and so never contradicts it (Appendix A.2).

The training objective sums the two bounds, with a weight ω on the intrinsic path and per-term weights on the divergences,

$$\begin{aligned} J = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} & \left[\frac{1}{\ell_i} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \mathbf{w}_i) \right. \\ & + \omega \frac{1}{\ell_i} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \check{\mathbf{w}}_i) \\ & - (1 + \omega) \alpha_z \text{KL} (q(\mathbf{z}_i) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) \\ & - \alpha_w \text{KL} (q(\mathbf{w}_i) \parallel \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I})) \Big] \\ & - \alpha_a \text{Pen}, \end{aligned} \quad (17)$$

where $\mathcal{S}$ is the set of cells whose loss is evaluated in a batch, $\check{\mathbf{w}}_i \sim p(\mathbf{w}_i \mid \mathbf{c}_i, t_i)$ is a draw from the prior, and Pen is the invariance term of Section 2.5, a functional of the whole batch; under the adversarial escalation it is replaced by the per-cell mean $\frac{1}{|\mathcal{S}|} \sum_i \text{Adv}_i$. One decoder is evaluated twice, with $\mathbf{w}$ from the posterior in the first reconstruction term and from the prior in the second. The factor $(1 + \omega)$ is what would keep J proportional to $L_i + \omega L_i^z$, at $\alpha_z = \alpha_w = 1/\bar{\ell}$ and $\alpha_a = 0$: each of Equations (15) and (16) carries its own $-\text{KL}(q(\mathbf{z}_i) \parallel p(\mathbf{z}_i))$, so their weighted sum contains $(1 + \omega)$ copies of it, as in Slavutsky et al. (2026, eq. 6), and omitting the second copy, the natural mistake, breaks the correspondence even at unit weights.

Scaling. The reconstruction terms are divided by ℓ_i . Unscaled, they grow in proportion to a cell’s depth and exceed the divergences by orders of magnitude, and depth differs widely between tissues, so weights tuned on one section would not transfer to another. The scaling changes what the weights mean: reconstruction becomes $O(1)$ while the divergences stay absolute, so $\alpha_z = 1$ prices a latent’s information roughly $\bar{\ell}$ -fold above the unscaled bound and collapses $\mathbf{z}$. At $\alpha_z = \alpha_w = 1/\bar{\ell}$, with $\bar{\ell}$ the mean total count, J is proportional to $L_i + \omega L_i^z$ for a cell of average depth,

and that point, not 1, is the centre of any sweep over the weights. We use $\alpha_z = \frac{1}{2}/\bar{\ell}$, fixed on a ladder over $\{\frac{1}{4}, \frac{1}{2}, 1, 2\} \times 1/\bar{\ell}$ read on the diagnostics of Section 2.7: because both bounds carry the $\mathbf{z}$ -divergence, the objective charges it $(1+\omega)$ -fold, and at $\omega = 1$ the half weight restores a single charge at the bound-equivalent point. A per-cell scaling against global weights also means that a low-count cell’s divergences are weighted more heavily relative to its likelihood than a high-count cell’s, pulling sparse cells toward the prior; we accept this as the price of transferable weights. Once any weight departs from $1/\bar{\ell}$, or through the per-cell scaling itself, J is a weighted surrogate rather than a bound; and because one $q(\mathbf{z})$ is shared by two variational families, q is a compromise between them, not the best posterior approximation under either, even at the bound-equivalent weights. The two reconstruction paths are bounds on the same evidence (Appendix A.2), but J , their scaled sum together with the invariance term, is a training criterion, and training does not maximise it as a single function: neighbour compositions enter under a stop-gradient, so each step treats the current $\bar{\rho}$ as data and ascends a partial gradient (Appendix A.1), and under the adversary the fit is a saddle point of a two-player game. We therefore claim no likelihood-based uncertainty for q ; the reported variability comes from the leakage sweep and the seeds. The hyperparameters are therefore ω , α_z , α_w and the response width d_w , plus α_a and, under escalation, the adversary’s update schedule; κ is not a hyperparameter but the sweep axis.

2.5 Conditional Invariance of $\mathbf{z}$

Architecture alone does not keep the environment out of $\mathbf{z}$. The counts $\mathbf{x}_i$ carry environmental signal, and with $d_z > d_w$ the intrinsic latent will absorb it given the chance. The target is

$$I(\mathbf{z}_i; [\mathbf{y}_i, \Phi_i] \mid t_i) = 0. \quad (18)$$

Two features of this target are deliberate. It is *conditional on type*, the equalised-odds form of an invariance constraint (Hardt et al., 2016; Zhang et al., 2018). The marginal constraint $I(\mathbf{z}; \mathbf{y}) = 0$ is the analogue of demographic parity, which must discard label information whenever label and protected variable are dependent (Zhao and Gordon, 2019). In tissue they are, because types cluster in space: carrying “I am a T cell” predicts “my neighbours are T cells”. A collapse of the code’s agreement with cell type under such an invariance, with the nuisance information driven to zero, is also documented for conditional subspace VAEs (Klys et al., 2018; Slavutsky et al., 2026). The cost of conditioning is that intrinsic variation that is spatially organised within a type is pushed into $\mathbf{w}$ to the extent that composition and image predict it, tumour subclones being

the sharp case; the decomposition is relative to t , not absolute, and lineage-level labels are preferable to finer or state-defined ones, for several reasons including a selection one (Appendix B.4). It also targets *both* $\mathbf{y}$ and Φ . If only composition were penalised, $\mathbf{z}$ would be free to absorb the morphology-defined niche (vessel proximity, fibrosis, density, necrosis), and $\mathbf{w}$ would under-report morphology-driven effects; $\mathbf{y}$ and Φ are correlated, so the part that matters is the residual that is image-predictable but not composition-predictable, and that residual is what would go unpenalised. Neither the penalty below nor the adversary is part of any bound; both are posterior regularisation (Ganchev et al., 2010), a constraint on the variational family written as a penalty, which is the status the adversarial term also has in Slavutsky et al. (2026) (Appendix A.3).

Closed-form penalty. The first regulariser is the Gaussian estimate of the conditional mutual information, which needs no second optimiser. With $\mathbf{v}_i = [\mathbf{y}_i^{(-K)}, \text{PC}(\Phi_i)]$, where $\mathbf{y}_i^{(-K)}$ drops one composition column (the simplex constraint makes the full block singular) and $\text{PC}(\Phi)$ keeps the leading principal components of the image embedding,

$$\text{Pen} = \sum_t p(t) \cdot \frac{1}{2} \left[\log \det \hat{\Sigma}_{\mathbf{v}|t} + \log \det \hat{\Sigma}_{\mathbf{z}|t} - \log \det \hat{\Sigma}_{[\mathbf{z}, \mathbf{v}]|t} \right], \quad (19)$$

evaluated on the posterior means $\boldsymbol{\mu}_z$ rather than on samples, since encoder noise would dilute the measured dependence and hiding dependence behind noise is exactly what the penalty must not permit (Appendix A.4). Three estimation details are not optional at realistic batch sizes. The image block is kept small (a dozen principal components), which keeps the joint block well conditioned and retains the components that carry the niche signal. The per-type moments are accumulated as exponential moving averages across batches (rate 0.05, an effective window of about twenty batches), because a spatial batch contains few cells of any rare type; a type enters the penalty only once its window-weighted cell count reaches ten times the joint dimension, and the fraction of cells excluded by this support floor is reported, since those cells go unregularised. The type weights $p(t)$ are the section-wide type frequencies and are not renormalised after an exclusion. Regularisation shrinks toward the diagonal (Ledoit and Wolf, 2004; Schäfer and Strimmer, 2005), $\hat{\Sigma} \leftarrow (1-\gamma)\hat{\Sigma} + \gamma \text{diag}(\hat{\Sigma})$, which damps only the correlations; an additive ridge $+\lambda \mathbf{I}$ would inflate every variance by an amount that depends on how ill-conditioned the estimate was, biasing the log-determinant upward most for rare types and weakening the penalty exactly where data are thinnest. Two further details concern

the gradient. The value of Equation (19) is taken from the moving-average moments but its gradient from the current batch at full scale, a straight-through estimator (Bengio et al., 2013); otherwise the only differentiable path runs through the averaging weight, and the effective strength of α_a would silently scale with the averaging constant. And the log-determinants are computed on correlation rather than covariance matrices: mutual information is invariant to per-dimension rescaling, so the value is identical, while the gradient of a log-determinant, which involves the inverse matrix, stays bounded when a dimension has near-zero variance, as the composition of cells deep inside one niche does. A jitter of $10^{-5}\mathbf{I}$ is added to the unit-diagonal correlation matrix before the factorisation; on a correlation matrix this is a uniform relative perturbation, not the variance-dependent ridge rejected above.

Adversarial escalation. The Gaussian penalty removes linear dependence only. When the probe below finds that dependence survives it, the penalty is replaced by an adversary (Louizos et al., 2016; Xie et al., 2017; Ganin et al., 2016; Zhang et al., 2018): two heads with parameters $\xi, \hat{\mathbf{y}}_\xi, \hat{e}_\xi^\Phi : (\mu_{z,i}, t_i) \mapsto \Delta^{K-1}, \Delta^{E_\Phi-1}$, trained by a separate optimiser to predict the composition and an image-niche membership from the posterior mean of the intrinsic code (evaluated on the mean for the same reason as Equation (19)), and an encoder term

$$\begin{aligned} \text{Adv}_i = & \text{CE}(\mathbf{y}_i, \bar{\mathbf{y}}(t_i)) - \text{CE}(\mathbf{y}_i, \hat{\mathbf{y}}_\xi(\mu_{z,i}, t_i)) \\ & + \text{CE}(e_i^\Phi, \bar{\Phi}(t_i)) - \text{CE}(e_i^\Phi, \hat{e}_\xi^\Phi(\mu_{z,i}, t_i)), \end{aligned} \quad (20)$$

which is the heads’ excess predictive skill over knowing the type alone, in categorical cross-entropy. The adversary maximises it, the encoder minimises it, and it is zero at the optimum; the two halves are reported separately. The image target $e_i^\Phi \in \Delta^{E_\Phi-1}$ is a soft cluster membership of Φ_i over E_Φ image niches, obtained by k -means on its leading principal components (never on the raw high-dimensional embedding, where distances are noise-dominated), fitted once and frozen. We set $E_\Phi = K$ so that the two heads face comparable difficulty and one weight α_a serves both; soft rather than hard assignments give a smoother target with no boundary artefacts. The adversary and the probe below therefore target the image through different summaries, discrete memberships and continuous components respectively; the memberships are the natural target for a classification head, the components for the Gaussian penalty and its probe. A weak adversary is worse than none: heads that are updated too slowly are defeated by the encoder while an independent probe still finds the leak, so head strength must be verified by the probe, never by the heads’ own training loss. This is the observation of Elazar and Goldberg (2018) that adversarial

removal is not enough without an independent check.

The probe, and how to decide. Whether the regulariser worked is decided by a *fresh* predictor of the invariance block $\mathbf{v}_i$ from $(\mu_{z,i}, t_i)$, trained on a spatially separate training split and evaluated on held-out cells: never the training adversary, and never the training cells. We report the excess held-out skill over the type-only baseline,

$$\Delta\text{CE} = \frac{1}{d_v} \left(\mathbb{E} \|\mathbf{v} - \bar{\mathbf{v}}(t)\|^2 - \mathbb{E} \|\mathbf{v} - \hat{\mathbf{v}}_{\text{probe}}(\mu_z, t)\|^2 \right), \quad (21)$$

the Gaussian form of the cross-entropy gain on the continuous block (mean squared error over its $d_v = K - 1 + 12$ components, which is the cross-entropy up to constants), with two reference scales that make it interpretable: the *noise floor*, the value obtained when μ_z is permuted across cells within type, and the *uncontrolled baseline*, the value from a model fit at $\alpha_a = 0$. The probe is fitted in two families, a ridge regression and a small multilayer perceptron (Table 3); the permutation floor is reported for the ridge probe, and the decision is taken on the nonlinear one, since the Gaussian penalty can leave nonlinear dependence in place by construction. The rule is to escalate from Equation (19) to Equation (20) when the nonlinear probe of a converged closed-form fit is clearly above the floor, and then to take the smallest α_a that brings the nonlinear probe within a small fraction of the uncontrolled baseline without a loss in the agreement between $\mathbf{z}$ and cell type or in held-out reconstruction. The agreement is the normalised mutual information (Strehl and Ghosh, 2002) between t and a k -means clustering of μ_z with $k = K$ clusters. If no such α_a exists, the invariance is costing cell type and the label set is too fine. **[TODO: state the fractions actually used once the calibration scans are repeated]**

2.6 Frozen Neighbours and Batching

Neighbour quantities are treated as data, not as things to back-propagate through. The neighbour compositions ρ_j in Equation (5) enter under a stop-gradient. The mechanism this prevents is a degeneracy rather than an explosion: with the path open, cell j ’s clean composition is trained to absorb cell i ’s residual through $\bar{\rho}_i$, so ρ_j stops being j ’s own profile, and the coupling feeds back through $\bar{\rho}_j$. Ergen et al. (2025) disable gradients on precisely this path after observing mode collapse with it enabled; a first-order feedback argument bounds the amplification through the coupling by $1/(1 - \kappa)$ (Appendix A.5). The neighbours’ responses $\mathbf{w}_j$ needed for ρ_j are posterior draws computed in the same batched pass and likewise placed under a stop-gradient.

A tempting shortcut is to compute the foreign influx from the baseline programme alone, $\rho_j \approx \text{softmax}(a(\mathbf{z}_j))$, which avoids the neighbours’ context entirely. It errs in the direction that matters most. Neighbours share niches, so $\mathbf{w}_j \approx \mathbf{w}_i$; dropping $\mathbf{w}_j$ strips the local spatial programme out of $\bar{\rho}_i$, and whatever the leakage term then fails to absorb is explained by $\mathbf{w}_i$ instead, *inflating* apparent spatial effects, the exact failure the leakage sweep exists to guard against.

Two hops, one pass. Because $\bar{\rho}_i$ needs ρ_j , hence $\mathbf{w}_j$, hence $\mathbf{c}_j$, hence $\mathcal{N}_j$, a training step for cell i reaches two graph hops even though the attention layer itself reaches one. Batching by *contiguous spatial tile* makes this cheap, because the halo of extra cells around a tile is shared rather than disjoint. A tile’s cells are the *seeds*; their graph neighbours form *ring one*, for which the context, the response posterior and the clean composition are all computed; the neighbours of ring one form *ring two*, for which only the type and the image embedding are looked up, since the attention sources are types. Both rings are built by look-ups in the same edge list that β uses, never by a distance query: a cell across a pruned edge is not in ring one even if a radius search would return it, and if the halo and the leak kernel disagree the foreign influx is silently wrong. Attention edges are restricted to destinations in the seeds and ring one, since a ring-two cell’s context would normalise over a truncated neighbourhood. The loss is evaluated on seeds only. With tiles of a few thousand cells the halo adds a modest fraction to the number of nodes processed per step, falling with tile size, cheaper than a cache of clean compositions that must be kept current; scattered seeds, by contrast, would cost many times more nodes than a contiguous tile, so spacing seeds apart is the expensive option, not the cheap one. The arithmetic of a pass is dominated by the encoder and decoder layers, at $O(Ngh)$ for hidden width h ; the foreign influx costs $O(|E|G)$ for $|E|$ directed edges, about $6NG$ at the Delaunay average degree, though its gathered edge-by-gene rows are the largest intermediate tensor of a step. One consequence of tile batching should be stated: the ring of a training tile can contain cells of a held-out tile, whose counts then enter training as inputs under a stop-gradient. No gradient flows from them and no loss is evaluated on them, but they are used transductively [**PENDING RUNS: quantify the fraction of training seeds with a held-out neighbour, or add a one-hop buffer around held-out tiles**].

2.7 Model Selection and Diagnostics

A model that explains everything with leakage reconstructs held-out cells well, so held-out reconstruction alone is not a valid stopping criterion. We stop on held-

out reconstruction and on the agreement between $\mathbf{z}$ and cell type *jointly*. The agreement is the normalised mutual information between t and a k -means clustering of μ_z (Section 2.5), computed on a random subsample of all cells; the clustering is unsupervised by design, so the quantity asks whether the geometry of $\mathbf{z}$ still separates the types rather than whether a classifier could recover them. Because the encoder receives the label (Section 2.2), it measures whether $\mathbf{z}$ *retains* type, not whether it discovers it, and its collapse is the detector for the failure mode in which the invariance machinery empties $\mathbf{z}$ while the remaining reads still look well behaved. The weight on the response divergence is not applied at full strength from the first step: the coefficient multiplying $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ is $\alpha_w \min(1, e/30)$ at epoch e , so it rises linearly from zero and equals α_w from epoch 30 on; no other term is scheduled. Without this ramp the conditional prior can fail to open on a given seed, the posterior of $\mathbf{w}$ collapsing onto a prior that ignores $\mathbf{c}$ within the first ten epochs and never recovering; the collapse is detected as a divergence that stays at zero and as zero mutual information between $\mathbf{w}$ and the niche on held-out cells, and a fit that shows it is discarded and re-seeded. The converged objective is unchanged. During the ramp the model is evaluated as usual but nothing is decided: no checkpoint is accepted, the patience count does not begin, and the running maximum of the type agreement is not updated. A checkpoint is accepted only if it improves held-out reconstruction while the agreement stands at or above a fixed fraction (0.9) of its running maximum over the fit. The maximum ratchets upward, so the bar rises as training finds a better $\mathbf{z}$, and a checkpoint that buys reconstruction at the cost of type structure is rejected outright rather than traded off. Held-out cells are entire spatial tiles, never random cells, because spatial autocorrelation leaks the answer through neighbouring cells landing on both sides of a random split (Roberts et al., 2017).

Budget and selection across seeds. Two budgets are used, and they are not interchangeable. Sweep points, where many fits must be comparable to each other rather than individually optimal, run a short budget (cosine annealing over 200 epochs, patience 20). Reference fits, the ones from which figures and single-fit analyses are taken, run a long budget (500 epochs, patience 40). The distinction matters because the objective is not unimodal in the seed: repeated fits of the same configuration at the long budget fall into more than one basin, and the better basin is not reached at the short budget at all. A reference model is therefore fitted from several random seeds and one member is selected, but never on held-out reconstruction alone: a fit can buy reconstruction while losing the intrinsic

state readouts, and at least one member of the family does. Selection is on the full diagnostic battery below, taken jointly — held-out reconstruction, the $\mathbf{z}$ -type agreement, the invariance probe, the mirror check, and the external-target readouts — with the remaining seeds reported as the envelope rather than discarded. Sweep trajectories are always read over their own seeds at the short budget and are not compared, point for point, with a long-budget reference fit.

Five diagnostics accompany every fit. (i) The within-type coefficient of determination of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$, against a control in which $\mathbf{c}_i$ is permuted among cells of the same type; a pooled version would partly measure type separability, which both variables legitimately carry, and the within-type form is blind to it. This within-type check complements the mirror diagnostics of Appendix B.1; Section 2.3 closes both routes by which $\mathbf{c}_i$ could echo $\mathbf{z}_i$, and the check is retained as verification. (ii) $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ per dimension. Its collapse to zero has three causes that the training curves alone cannot distinguish: posterior collapse under a large α_w , the mirror, and a response posterior starved of ego evidence. Each has its own check and fix (Appendix B.3), and the mirror must be ruled out before any free-bits style remedy (Kingma et al., 2016) is applied, since free bits would mask it and leave the anomaly score meaningless. (iii) The held-out probe of Equation (21). (iv) The $\mathbf{z}$ -type agreement, the hard floor whose collapse is the marginal-invariance failure mode. (v) Two degeneracy checks in the opposite direction, asking whether $\mathbf{z}$ has collapsed onto the label: the ratio $I(\mathbf{z}; t)/H(t)$ together with the within-type variance of μ_z , and the gap in held-out reconstruction between $\mathbf{z}$ and a one-hot type code [PENDING RUNS: compute for the canonical fits].

2.8 What Is Identified, and the Leakage Sweep

Identifiability. Four statements can be made. The leak fraction κ is not identified from the count matrix under the parametric forms we are willing to assume, and is swept. Given κ , the response coordinates are identified only up to a rotation: the isotropic prior of Equation (3) and the linear decoder are invariant under $\mathbf{w} \mapsto \mathbf{Q}\mathbf{w}$, $\mathbf{B} \mapsto \mathbf{B}\mathbf{Q}^\top$, $m_\psi \mapsto \mathbf{Q}m_\psi$ for orthogonal $\mathbf{Q}$ (the learned prior mean absorbs the last), while the fixed prior scale removes the rescaling freedom (Section 2.3); the diagonal posterior of Equation (14) is not rotation-equivariant, so this is a symmetry of the model, not of the fitted objective. Through the softmax, each column of $\mathbf{B}$ is further identified only up to an additive constant over genes, and so is $a(\mathbf{z})$ per cell; loadings are read only after centring each column over genes. A second, less obvious freedom is a *per-*

type translation of the response. For any collection of constants $\mu_t \in \mathbb{R}^{d_w}$, one per type, the model $(a(\mathbf{z}) - \mathbf{B}\mu_t, \mathbf{w} + \mu_t)$ is the same model: the prior mean m_ψ takes t as an input and absorbs the shift, the divergence of Equation (14) is taken against that prior and is unchanged, and the foreign influx is built from the shifted response and the compensated baseline together. Nothing in the objective pins where the type offset lands, so where it lands is decided by the optimisation dynamics; parameter weight decay does not fix it, because $\mathbf{w}$ is a network output and not a parameter, and penalising parameters only moves the offset further into $\mathbf{w}$. Two consequences are carried through the evaluation. The magnitude $\|\mathbf{w}_i\|$, and any ranking of types by it, is a statement about the gauge and not about how strongly a type responds to its context; the gauge-free quantity is the spread of $\mathbf{w}$ *within* a type, or equivalently the realised shift $\mathbf{B}[m_\psi(\mathbf{c}_i, t_i) - m_\psi(\bar{\mathbf{c}}_t, t_i)]$ relative to a reference context $\bar{\mathbf{c}}_t$. And a “clean” per-cell profile must be decoded at such a reference context rather than at $\mathbf{w} = \mathbf{0}$, which would silently subtract the type offset along with the context effect. Differences between two contexts, which is what the counterfactual reads, are invariant to the gauge and need no correction. The allocation between $\mathbf{z}$ and $\mathbf{w}$ is not pinned by the likelihood, since both posteriors see the counts; it rests on the conditional-invariance constraint, on the intrinsic path, and on $d_w < d_z$. For $\mathbf{w}$ given the context this is in the spirit of the auxiliary-variable setting of Khemakhem et al. (2020), whose guarantee (recovery up to an affine map, under an injective additive-noise decoder and sufficient variability of the conditional prior) we do not establish for this likelihood; for $\mathbf{z}$ no comparable guarantee exists without the constraint (Locatello et al., 2019), and the constraint is what we verify with the probe rather than assume.

The sweep. The unit of inference is a sensitivity analysis with κ as the sensitivity parameter, in the spirit of partial identification (Manski, 2003; Rosenbaum, 2002). The model is fitted independently at each $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ with several random seeds per grid point, with every other weight held at values calibrated once at a reference grid point and the held-out probe checked at every κ so that a change in a readout is not a change in the invariance. Every readout of interest, the loadings $\mathbf{B}$, the per-type magnitude of the response, the spatial autocorrelation of each latent, the held-out probes and the agreement with external labels, is reported as a trajectory over κ with its envelope over seeds. We call a readout stable when its sign and, for rankings, its order are unchanged over the grid and its seed envelope excludes zero at every grid point [TODO: confirm this rule]. A

stable readout is a finding; one that changes sign, vanishes, or is inconsistent across seeds is not separable from leakage on this data. The sweep separates spatial association from leakage. It does not separate a response from intrinsic variation that is spatially organised within a type, which conditional invariance assigns to $\mathbf{w}$, nor from cells selecting their niche. Some readouts are expected to move with κ by construction: as κ grows, more of the resemblance between neighbours is assigned to the mixture and the fitted response magnitude shrinks, so a decline in $\|\mathbf{w}\|$ along the grid is mechanical, whereas a change in the *direction* of a programme or in which latent a target aligns with is not. Two consequences follow. Gene programmes are identified only up to the rotation and the per-column constants above, so raw loadings and probe directions are never compared across runs or seeds; only run-internal, rotation-invariant summaries are reported, and any gene-level statement is treated as relative to the run that produced it. And the reported range is not a confidence interval: it does not shrink with more cells, because the confound it expresses is not a sampling problem. An operating point on the grid is used for figures and for the analyses that require a single fit, and is always reported alongside the sweep.

What $\mathbf{w}$ is not. The context $\mathbf{c}_i$ is the environment: deterministic, shared by co-located cells, a conditioning variable. If a niche descriptor were all that was wanted, one would drop the latent and feed $\mathbf{c}_i$ to the decoder directly, which is roughly the node-centric model of Fischer et al. (2023) and is simpler. The response $\mathbf{w}_i$ is latent and per cell, and it earns its place only through within-niche heterogeneity: anomalies, per-cell responses, and counterfactuals with abduction. Whether it does so at a given operating point is an empirical question that the per-dimension divergence answers, and it is reported as such.

3 EXPERIMENTS

[TODO: Deferred. The model is still under development; every evaluation and experiment is added once the protocol is final. The drafted experiments section and the calibration, data, synthetic, validation and additional-results appendices are kept on disk, out of the build.]

4 CONCLUSION

[TODO: Deliberately empty for now.]

Future work. [TODO: Perturbations as a third source of variation: with a perturbed and an unperturbed condition the model gains a third

variable, and the decomposition becomes intrinsic state, microenvironmental response and perturbation response, with the leakage sweep applied to all three. Per-cell anomaly scores and counterfactuals with abduction once an operating point with a non-degenerate response posterior is established. Estimation of κ from a nuclear/extranuclear split of the counts.]

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Checklist

[TODO: Answer at submission time. Questions must not be modified; the instruction paragraph of the template has been removed.]

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [TODO: answer]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes]
 - (b) Complete proofs of all theoretical results. [Yes]
 - (c) Clear explanations of any assumptions. [Yes]
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [TODO: answer]
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [TODO: answer]
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. [TODO: answer]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

DISCELL: Supplementary Materials

A DERIVATIONS

A.1 The Reconstruction Bound

Fix a cell i and hold $\mathbf{c}_i$ and $\bar{\rho}_i$ constant, as Section 2.6 does; every probability in this appendix is conditional on $\bar{\rho}_i$, which we suppress from the notation. Then

$$\log p(\mathbf{x}_i | \mathbf{c}_i, t_i) = \log \int \int p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) p(\mathbf{w} | \mathbf{c}_i, t_i) p(\mathbf{z}) d\mathbf{z} d\mathbf{w} \quad (22)$$

$$\geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{q(\mathbf{w}|\mathbf{z})} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) + \log p(\mathbf{w} | \mathbf{c}_i, t_i) + \log p(\mathbf{z}) - \log q(\mathbf{z}) - \log q(\mathbf{w} | \mathbf{z}) \right] \quad (23)$$

$$= \mathbb{E}_{q(\mathbf{z})} \left[\mathbb{E}_{q(\mathbf{w}|\mathbf{z})} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) \right] - \text{KL}(q(\mathbf{w} | \mathbf{z}) \| p(\mathbf{w} | \mathbf{c}_i, t_i)) \right] - \text{KL}(q(\mathbf{z}) \| p(\mathbf{z})), \quad (24)$$

where Equation (23) is Jensen's inequality under the variational family $q(\mathbf{z}) q(\mathbf{w} | \mathbf{c}_i, t_i, \mathbf{z}, \mathbf{x}_i)$. Because $q(\mathbf{w} | \cdot)$ conditions on the sampled $\mathbf{z}$, the exact bound keeps the $\mathbf{w}$ -divergence inside $\mathbb{E}_{q(\mathbf{z})}$. The divergence between two diagonal Gaussians is available in closed form,

$$\text{KL}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \| \mathcal{N}(\mathbf{m}, \mathbf{s}^2)) = \sum_k \left[\log \frac{s_k}{\sigma_k} + \frac{\sigma_k^2 + (\mu_k - m_k)^2}{2s_k^2} - \frac{1}{2} \right], \quad (25)$$

and evaluating it at the single reparameterised draw of $\mathbf{z}$ used for the reconstruction term is an unbiased one-sample estimate of $\mathbb{E}_{q(\mathbf{z})}[\text{KL}(\cdot)]$.

Summed over cells, with each neighbour's $\mathbf{z}_j, \mathbf{w}_j$ drawn from its own posterior as in Section 2.6, $\sum_i L_i$ is a one-sample estimate of the evidence lower bound of the coupled model, $\log p(\{\mathbf{x}_i\} | \{\mathbf{c}_i, t_i\})$, under the fully factorised family $\prod_i q(\mathbf{z}_i) q(\mathbf{w}_i | \mathbf{z}_i)$. The per-cell bounds are not independent terms, since $\bar{\rho}_i$ is a function of the neighbours' latents. The stop-gradient removes the terms $i \neq j$ from the gradient with respect to q_j , so training ascends a partial gradient of this bound: a fixed-point iteration in which each step treats the current $\bar{\rho}$ as data.

A.2 The Intrinsic Path and the Factor $(1 + \omega)$

A second bound on the same evidence holds by restricting the variational family to $q(\mathbf{w}) := p(\mathbf{w} | \mathbf{c}_i, t_i)$, which zeroes the $\mathbf{w}$ -divergence and leaves

$$\log p(\mathbf{x}_i | \mathbf{c}_i, t_i) \geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{p(\mathbf{w}|\mathbf{c}_i, t_i)} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) \right] - \text{KL}(q(\mathbf{z}) \| p(\mathbf{z})) =: L_i^z. \quad (26)$$

Call the bound of Appendix A.1 L_i . Each carries its own $-\text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$, so $L_i + \omega L_i^z$ contains $-(1 + \omega) \text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$. Omitting the second copy breaks the correspondence with $L_i + \omega L_i^z$ even at $\omega = 1$.

The construction mirrors the two-bound objective of Slavutsky et al. (2026), where the sum of the shared-only and joint bounds corresponds to minimising $\text{KL}(q(\mathbf{z}) \| p(\mathbf{z} | \mathbf{x})) + \text{KL}(q(\mathbf{z})q(\mathbf{w}) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}, u))$ with u their condition label, a sum of two non-negative divergences. Here, unlike in Slavutsky et al. (2026), whose shared-only term decodes from $\mathbf{z}$ alone and so bounds a different model, both terms bound the same evidence,

$$L_i + \omega L_i^z \leq (1 + \omega) \log p(\mathbf{x}_i | \mathbf{c}_i, t_i),$$

with gap $\text{KL}(q(\mathbf{z})q(\mathbf{w} | \mathbf{z}) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}_i, \mathbf{c}_i, t_i)) + \omega \text{KL}(q(\mathbf{z})p(\mathbf{w} | \mathbf{c}_i, t_i) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}_i, \mathbf{c}_i, t_i))$. One $q(\mathbf{z})$ serves two variational families, so its optimum is a compromise and not the best approximation of the posterior under either, which is one of the reasons uncertainty is reported from the leakage sweep rather than from q . The invariance term sits outside the bound for a separate reason (Appendix A.3).

Why the intrinsic path is load-bearing: in Slavutsky et al. (2026) the shared code is kept informative by a reconstruction from $\mathbf{z}$ alone and by a prior on $\mathbf{w}$ centred on the class-wise means of $\mathbf{z}$. Our intrinsic path is the analogue of the first. We have no analogue of the second: our prior on $\mathbf{w}$ never references $\mathbf{z}$, and the prior on $\mathbf{z}$ is a standard normal, so nothing else in the objective rewards $\mathbf{z}$ for being informative. The intrinsic path supplies that pressure directly, since under it $\mathbf{z}$ must explain the cell given only the expected response. It runs through the same leakage mixture as the first term, so it never asserts $\kappa = 0$ and contradicts it.

A.3 The Invariance Term as Posterior Regularisation

Neither Pen nor Adv_{*i*} is part of any bound. Both are posterior regularisation in the sense of Ganchev et al. (2010): a constraint on the variational family, $I(\mathbf{z}; [\mathbf{y}, \Phi] | t) = 0$, written as a penalty on the variational objective. This is the same status Slavutsky et al. (2026) give their adversarial term. It is a statement about which q are admissible, not a modification of the generative model, and the generative model’s conditional independence $\mathbf{z}_i \perp [\mathbf{y}_i, \Phi_i] | t_i$ is what it encodes.

A.4 The Gaussian Conditional Mutual Information

For jointly Gaussian $(\mathbf{z}, \mathbf{v})$ with covariance $\Sigma_{[\mathbf{z}, \mathbf{v}]}$ and marginal covariances $\Sigma_{\mathbf{z}}, \Sigma_{\mathbf{v}}$, the mutual information is $I(\mathbf{z}; \mathbf{v}) = \frac{1}{2}[\log \det \Sigma_{\mathbf{z}} + \log \det \Sigma_{\mathbf{v}} - \log \det \Sigma_{[\mathbf{z}, \mathbf{v}]}]$ (Cover and Thomas, 2006). Equation (19) applies this per type and averages with the type frequencies, which is the Gaussian estimate of $I(\mathbf{z}; \mathbf{v} | t)$ under a type-wise Gaussian assumption. Two properties used in Section 2.5 follow directly. First, the value is invariant to a per-dimension rescaling $\mathbf{u} \mapsto \mathbf{D}\mathbf{u}$ with $\mathbf{D}$ diagonal: each log-determinant gains $\log \det$ of the corresponding block of $\mathbf{D}^2$, and the blocks of $\mathbf{D}$ for $\mathbf{z}$ and $\mathbf{v}$ together make up $\mathbf{D}$, so the three terms cancel exactly. The penalty may therefore be computed on correlation matrices. Second, on the simplex $\sum_k y_k = 1$ the covariance of the full composition vector is singular and the log-determinant is $-\infty$; dropping one column removes the constraint without changing the information, since the dropped column is a deterministic function of the others.

A.5 Feedback Amplification Through the Leakage Path (Heuristic)

The following is a first-order heuristic, not a theorem. Write the decoder output of cell j as ρ_j and suppose gradients were allowed to flow from cell i ’s reconstruction through $\bar{\rho}_i = \sum_j \beta_{ij} \rho_j$ into the parameters that produce ρ_j . Linearising around a fixed point of the coupled system, a perturbation δ of a cell’s clean composition induces a perturbation $\kappa \beta \delta$ of its neighbours’ fitted compositions, which feeds back through the same path; the total response is $\delta \sum_{n \geq 0} (\kappa \mathbf{P}_\beta)^n$ for a row-stochastic $\mathbf{P}_\beta$ built from β , whose spectral radius is at most one, so the geometric series is bounded by $1/(1 - \kappa)$. The stop-gradient truncates the series at $n = 0$. The argument concerns perturbations of the fitted compositions; the gradient fixed point through shared decoder parameters is more involved, and the empirical fact that matters is the collapse reported by Ergen et al. (2025) when the path is left open. **[TODO: either complete the gradient-level argument or keep this as a heuristic]**

B DESIGN RATIONALE

This appendix carries the parts of the arguments of Section 2 that the main text only summarises. It does not repeat what is already there.

B.1 The Mirror Attractor

The attention layer of Equation (10) computes logits $e_{ij} = \mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}_{\text{dst}} \mathbf{h}_i + \mathbf{W}_{\text{src}} \mathbf{h}_j)$ and aggregates $\sum_j \pi_{ij} \mathbf{W}_{\text{src}} \mathbf{h}_j$, so the destination feature $\mathbf{h}_i$ enters the *attention* but never the *value* (Brody et al., 2022). With $\mathbf{z}_i$ as the destination feature, the selection channel of Section 2.3 opens. What the type query gives up is state-specific attention, which belongs in the decoder (as type-dependent loadings or a low-rank interaction with $\mathbf{z}$) rather than in the context encoder.

The diagnostic is the within-type coefficient of determination of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$ against a within-type permuted control (Section 2.7). The correlation between π_{ij} and $\text{sim}(\mathbf{z}_i, \mathbf{z}_j)$ would serve under a $\mathbf{z}$ query; with the type as query and types as sources the attention weights depend on the types alone (Equation (27)), and the correlation is vacuous. Downstream, a mirror is indistinguishable from ordinary posterior collapse of $\mathbf{w}$; a free-bits remedy

Table 2: Segmented-polygon contact as a leakage kernel, against the Voronoi-face kernel β , on every section and on the model’s edge set (Delaunay adjacency pruned at $40\mu\text{m}$). Both kernels use the same distance decay $\exp(-d_{ij}/\tau)$ and the same row normalisation. The contact kernel weights an edge by the length of membrane along which the two segmented polygons run within $1\mu\text{m}$ of each other; β weights it by the Voronoi face. *Touching*: adjacent pairs whose polygons are at zero distance. *No-leak cells*: connected cells whose kernel row is all zero, so that they receive no leakage at any κ ; *sparse* and *dense* are the lowest and highest fifth of cells by the number of centroids within $20\mu\text{m}$. ρ : Spearman correlation, over edges, between a neighbour’s share of the row and the gap between the two polygons. The Voronoi kernel has no zero-weight edges and no no-leak cells on any section. Its dependence on the gap is that of the distance decay alone (shares from the decay alone: ρ between -0.49 and -0.62).

Section	Segmented by (% cells)			Edges (%)		No-leak cells, contact (%)			ρ (share, gap)	
	interior	boundary	nucleus	touching	no contact	all	sparse	dense	contact	β
Ovarian cancer (FFPE)	69	29	2.0	3.5	47	9.8	33.6	0.6	-0.82	-0.48
Lung cancer (FFPE)	82	16	2.2	2.7	57	11.1	27.4	0.7	-0.84	-0.55
Ovary (fresh frozen)	70	28	2.5	6.7	38	3.2	10.6	0.1	-0.80	-0.52
Lung TMA, section A	87	12	1.0	2.2	47	6.3	24.8	0.1	-0.84	-0.54
Lung TMA, section B	87	12	1.2	2.2	47	6.2	24.6	0.1	-0.84	-0.53

(Kingma et al., 2016) would mask it and leave the anomaly score meaningless, which is why the mirror is ruled out first.

Zeroing $\mathbf{z}_i$ while keeping self-loops is not an alternative: the softmax still spends attention mass on the self-edge, down-weighting neighbours, and under $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ the origin is the prior mean, so a zero vector asserts “this cell is average” rather than “ignore this cell”.

B.2 Why the Context Gets No Per-Edge Geometry, and Why the Graph Is Delaunay

A discretised contact indicator (touching, near, far) looks as if it dodges the collinearity objection of Section 2.3, but on this platform, whose segmented polygons rarely touch, whether two of them touch is largely a measurement of local density: dense regions touch, sparse regions do not. The indicator would smuggle density back in, in the confounded direction, and a tolerance parameter only moves the threshold.

Delaunay adjacency resolves it: two cells are neighbours if and only if their Voronoi cells share a face, which is the correct notion of adjacency for a packing, built from centroids alone. The one parameter that returns is the pruning of long edges, and it is much less sensitive than choosing a neighbour count. The Voronoi face length is used in the kernel rather than shared segmented boundary because the Voronoi face is where the boundary between two cells effectively lies, its length is a contact-extent proxy defined whether or not the polygons touch, and it is independent of the expansion setting.

Table 2 makes the comparison concrete on every section. The sections were segmented mostly by an interior RNA stain rather than by nuclear expansion, and adjacent polygons still rarely touch. A contact kernel is therefore zero on about half of the edges, and it leaves a density-dependent share of cells with no leakage at all: up to a third of cells in the sparsest fifth of a section, almost none in the densest. Whether a cell can receive leakage would then be decided by local density, which is part of the niche. The Voronoi kernel gives every connected cell the same leak budget κ . Both kernels give nearer neighbours more weight; for β that dependence is the one the distance decay introduces by design, and the face does not add to it.

One consequence of the softmax over a Delaunay neighbourhood should be stated. With the type as query and one-hot types as sources, the logit of edge ij depends on (t_i, t_j) alone, say $s_{t_i t_j}$, and grouping the neighbours by type gives, per head,

$$\mathbf{g}_i = \frac{\sum_k y_{ik} e^{s_{t_i k}} \mathbf{W}_{\text{src}} \mathbf{e}_k}{\sum_k y_{ik} e^{s_{t_i k}}}, \quad (27)$$

with $\mathbf{e}_k$ the k -th unit vector: a learned, type-specific reweighting of the neighbour composition in which the degree cancels. The context therefore depends on the fractions $\mathbf{y}_i$, not on the counts. It distinguishes one tumour neighbour among six from five among six, but not a neighbourhood from the same neighbourhood with every type count scaled up together; in particular a single-type neighbourhood gives the same context whatever its

size. On this tessellation the degree is nearly constant across the section, so a fraction determines a count and the blindness costs little. An attention sink, a fixed null logit that joins every softmax and restores a saturating dependence on the count, is implemented and was tested over three seeds. It learned no dose response, placing its half-point below a single neighbour, did not improve the held-out transport of the response, and in two of the seeds made the prior mean read the context less faithfully. It is therefore left off. It would become meaningful only on a graph whose degree varies, where the invariance target would have to grow with it.

The fallback is empirical. If m_ψ predicts responses poorly, or held-out reconstruction is worse than a comparison run whose attention layer carries edge features (distance, face length), that is direct evidence that the continuous signal mattered. If it is added back, the correlation between π_{ij} and β_{ij} must be monitored: a high correlation means the attention has rediscovered the leakage kernel and $\mathbf{w}$ is becoming collinear with $\bar{\rho}$. That risk is tolerable while κ is fixed, since there is no free contamination parameter to feed, and intolerable once κ is estimated. **[PENDING RUNS: the edge-feature comparison run has not been performed]**

B.3 Three Routes to a Vanishing Response Divergence

Cause	Check	Fix
Posterior collapse	α_w large relative to reconstruction	lower α_w ; free bits
Mirror	within-type R^2 of $\mathbf{z}_i$ on $\mathbf{c}_i$ vs. permuted control	type embedding as query; types only as sources
Starved of ego evidence	does the response posterior receive $\mathbf{x}_i$ directly?	add the direct path

B.4 Conditional Invariance and Label Granularity

The marginal-invariance failure is documented by Slavutsky et al. (2026) for the conditional subspace VAE (Klys et al., 2018), whose invariance drives the agreement between the shared code and the nuisance variable to zero while collapsing its agreement with cell type **[TODO: quote the exact numbers from that comparison]**. Lineage-level labels are preferable to finer or state-defined ones for four reasons: fine labels leave $\mathbf{z}$ with nothing to carry, they block more of the position→fate→expression path, they destroy the overlap that any conditional comparison needs, and, because t_i is itself a function of $\mathbf{x}_i$, hence of leakage and of any response that moves a cell across a label boundary, conditioning on it conditions on a descendant of the niche, so that even the true intrinsic state can depend on $\mathbf{y}_i$ given t_i near label boundaries (Hernán et al., 2004).

B.5 The Leak Fraction: Grid Range and Upgrade Path

The grid range comes from three sources: zero as anchor; published misassignment estimates for this platform **[NEEDS A SOURCE]**; and a marker-set ceiling. For an atlas-defined set $\mathcal{M}$ of genes that a cell of type t_i should not express, and an empirical neighbour profile $\tilde{\rho}_i$ obtained by averaging the observed compositions of i 's neighbours with the kernel β , the ratio $\sum_{g \in \mathcal{M}} x_{ig} / (\ell_i \sum_{g \in \mathcal{M}} \tilde{\rho}_{ig})$ bounds the leak fraction from above. The set must be defined externally: ranking genes by their neighbour-to-own expression ratio from the same counts selects the zero-count genes and drives the bound to zero with probability one. **[PENDING RUNS: compute the marker-set ceiling on both sections]**

This version of the model sweeps κ ; it cannot estimate it. Assigning transcripts to cells by position does not settle it either: counts of transcripts that lie nearer a neighbour's nucleus than the cell's own are dominated by segmentation geometry, since a large cell beside small ones counts its own far-side transcripts as foreign, so such counts bound leakage from above but do not identify how it scales with the depth of the receiving or the donating cell. Estimation needs two views of one cell that share a foreign profile at different contamination levels, which a nuclear/extranuclear split of the transcript coordinates provides. Its identifying condition is that gene-wise nuclear retention varies across genes, which is also a reason to expect the gene-specific leakage that the main model does not represent (Section 2.2), and which a two-view likelihood would be the natural place to model; with constant retention both views lie on the same line between ρ_i and $\bar{\rho}_i$ and nothing is pinned. Adding it does not restructure the model: the likelihood becomes a multinomial over $2G$ bins and κ moves from hyperparameter to latent. We leave it as future work.

B.6 Learned Variances: What to Learn and What Not To

If a conditional $\sigma_w(\mathbf{c}, t)$ is wanted, because it would standardise deviation against local variability so that a deviation of two in a niche where everyone deviates by two does not read as anomalous, it must be bounded, e.g. $\sigma_w \in [0.5, 2]$ through a scaled sigmoid, so that it cannot inflate away, and its mean over cells must be checked for upward drift over training.

The prior on $\mathbf{z}$ is a standard normal. A richer prior, a mixture with one component per type or a VampPrior with archetypal pseudo-inputs (Tomczak and Welling, 2018), would be a genuine upgrade for cell-type structure and would make t a latent rather than an input, but that is a different change, not a variance question.

C IMPLEMENTATION DETAILS

C.1 Architecture and Hyperparameters

Table 3 lists every network and every fixed quantity as implemented; the weights marked calibrated are set per section by the protocol of Section 2.5 and are not quoted here. All multilayer perceptrons use rectified-linear hidden units.

C.2 Choices Made in Implementation

Where the model description of Section 2 leaves a choice open, Table 4 records what was done and why. Every entry is also stated in the method text; the table collects them in one place. **[TODO: author to decide whether this table stays; it duplicates the method text]**

Table 3: Networks and fixed quantities. G genes, K types, d_Φ image-embedding dimension.

Component	Input	Layers / value
Count encoding $q(\mathbf{z}_i \mathbf{x}_i, t_i)$	$\mathbf{x}_i$ count encoding $\oplus$ onehot(t_i)	$[\log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i), \log \ell_i]$, $\tilde{\ell}$ the median total $256 \rightarrow 256 \rightarrow 2d_z$, $d_z = 20$
Type query embed(t_i)	t_i	learned table, width K
Attention sources	neighbour j	onehot(t_j), width K ; an earlier version appended the neighbour's posterior mean under a stop-gradient and was dropped after a three-seed ablation
Attention layer	sources, query	one bipartite GATv2 layer, 4 heads $\times$ 32, averaged, LeakyReLU slope 0.2, no self-loops, no edge features
Context $\mathbf{c}_i$		attention output (32) $\oplus \Phi_i$ ($d_\Phi = 384$) $\oplus$ isolation flag (1)
Prior mean $m_\psi(\mathbf{c}_i, t_i)$	$\mathbf{c}_i \oplus$ onehot(t_i)	$64 \rightarrow d_w$, $d_w = 6$
$q(\mathbf{w}_i \cdot)$	$\mathbf{c}_i \oplus$ onehot(t_i) $\oplus \mathbf{z}_i \oplus$ count encoding	$256 \rightarrow 2d_w$
Decoder $a(\mathbf{z}_i)$	$\mathbf{z}_i$	$256 \rightarrow G$
Loadings $\mathbf{B}$	$\mathbf{w}_i$	linear $d_w \rightarrow G$, no bias
Adversary heads	$\mu_{z,i} \oplus$ onehot(t_i)	two heads $64 \rightarrow 64 \rightarrow K$ (composition) and $\rightarrow E_\Phi = K$ (image niche)
Image niche e_i^Φ	12 principal components of Φ	soft k -means memberships softmax($-d^2/T$), T the median nearest squared distance; the projection is fitted on a 50,000-cell random subsample and applied to all cells, the k -means (four restarts) on all cells; fitted once and frozen
Invariance block $\mathbf{v}_i$		$[\mathbf{y}_i^{(-K)}, 12 \text{ principal components of } \Phi_i]$
Held-out probes	$\mu_{z,i} \oplus$ onehot(t_i)	ridge regression ($\lambda = 10^{-3}$) with a within-type permutation floor; perceptron $64 \rightarrow 64$, 300 Adam steps at 10^{-3} with batches of 2,048; both fitted on at most 30,000 training-tile cells
Graph		Delaunay adjacency; Voronoi faces measured within $30 \mu\text{m}$ discs; prune at $40 \mu\text{m}$
Kernel β		$f_{ij} \exp(-d_{ij}/\tau)$, $\tau = 20 \mu\text{m}$, row-normalised over in-edges
Prior scale σ_w		1 (fixed)
Leak fraction κ		swept over $\{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$; the operating point is a grid point chosen on the primary section
ω		1
α_z		$\frac{1}{2}/\bar{\ell}$, half the bound-equivalent value (Section 2.4), set per section from its mean total count
α_w		calibrated
α_a (adversary)		calibrated; the number of head updates per model update and the head learning rate are set so that the independent probe, not the heads' own loss, certifies head strength
α_a (closed form)		calibrated by the same probe protocol
Moving-average shrinkage	rate,	0.05, 0.05; support floor $10 \times$ joint dimension
Tiles		recursive median split into rectangles of $\leq 4,096$ cells; 15% of tiles held out
Optimiser		Adam (Kingma and Ba, 2015), learning rate 10^{-3} , cosine annealing (Loshchilov and Hutter, 2017) over the epoch budget, gradient-norm clip 10
Evaluation		every 5 epochs on posterior means; checkpoint accepted if held-out reconstruction improves and $\mathbf{z}$ -type agreement $\geq 0.9 \times$ its running maximum; patience in epochs set with the budget; α_w ramped linearly from 0 over the first 30 epochs, during which no checkpoint is accepted and patience does not count
Hardware		one consumer GPU

Table 4: Implementation choices and their reasons.

Where	Decision	Reason
Type input to $q(\mathbf{w})$ and m_ψ	one-hot everywhere the type is an input; the learned type embedding exists only as the attention query, sized to match the source features	the query stands in for the $\mathbf{z}_i$ that the cell deliberately does not contribute to its own context
Counts input to $q(\mathbf{w})$	the same encoding as $q(\mathbf{z})$	one encoding of the counts for both posteriors
Neighbour responses $\mathbf{w}_j$	a posterior draw from the same batched pass, under a stop-gradient	the neighbours' clean compositions must include their local programme (Section 2.6)
Invariance block	one composition column dropped	the simplex makes the full block singular; the information is unchanged (Appendix A.4)
Penalty matrices	correlation rather than covariance form, plus $10^{-5}\mathbf{I}$ on the correlation matrix before the log-determinant	identical value by scale invariance; bounded gradient; on a unit-diagonal matrix the additive term is a uniform relative jitter, not a variance-inflating ridge
Penalty gradient	straight-through: value from the moving averages, gradient from the batch	otherwise the effective strength of α_a scales with the averaging rate
Penalty, adversary and probe input	posterior means $\boldsymbol{\mu}_z$, not samples	encoder noise would dilute the measured dependence
Objective granularity	per-cell mean over the batch rather than a sum	invariant to tile size
Evaluation draws	posterior means	the stopping signal should not ride reparameterisation noise
Image embedding dimension	the full native embedding into $\mathbf{c}_i$ rather than a 32–64-dimensional projection	the deciding test is held-out reconstruction with and without the projection; the invariance block uses 12 principal components regardless
Probe target for the image	Gaussian cross-entropy on the continuous block rather than categorical cross-entropy against image-niche clusters	the closed-form penalty takes continuous components directly; a small nonlinear probe is fitted alongside as a cross-check
Graph construction	the Voronoi tessellation is clipped to $30\mu\text{m}$ discs before faces are measured, then edges are pruned at $40\mu\text{m}$	the clip removes faces only for pairs more than roughly $60\mu\text{m}$ apart, which the prune removes anyway, and bounds the length of faces at tissue borders and across gaps; the effect on β for kept edges has not been quantified [TODO: quantify]
Invariance mechanism	the adversary is the operating configuration; the closed-form penalty is the cheaper first attempt	the escalation rule of Section 2.5 decides between them on each section

Table 5: Implementation choices, continued: the configuration used in the experiments, and what exists but is switched off.

Where	Decision	Reason
Attention sources	neighbour types only; an earlier version also passed the neighbour’s posterior mean under a stop-gradient	the case is empirical, not mechanistic: a seed-matched ablation improved held-out reconstruction, the $\mathbf{z}$ -type agreement, the mirror check and the cycle readout of $\mathbf{w}$ while leaving $\mathbf{w}$ ’s spatial readouts unchanged. The mechanism originally proposed for it, the prior reconstructing a cell from its neighbours’ codes, was tested with two purpose-built detectors and not confirmed; the sources are types because the ablation is better, not because that mechanism was demonstrated
Training budget	two budgets: a short one for sweep points, a long one for reference fits (Section 2.7)	sweep points must be comparable to one another; reference fits must be individually good, and the better optimisation basin is not reached at the short budget
Choosing among seeds	several seeds per reference configuration, one selected on the full diagnostic battery, the rest reported as the envelope	selection on held-out reconstruction alone admits a fit that buys likelihood at the cost of the intrinsic-state readouts
Halo overhead	ring one is fully encoded; ring two contributes a type and an image lookup only	the attention sources are types, so ring two needs no encoder pass. The implementation as run still evaluates the count encoder on ring two and discards the result, so the measured node overhead is an upper bound on what the design requires; it falls with tile size, and tiles are sized above the knee
Resident count storage	the count matrix is held on the accelerator as 16-bit integers and cast per tile	counts are small non-negative integers, so the narrow type is exact and is asserted to be; the wide type does not fit on one consumer card for the largest section, and tile size does not help, since every tile is resident
Built but not default	leak-subtracted encoder input $\tilde{\mathbf{x}}_i = \max(\mathbf{x}_i - \kappa \ell_i \tilde{\boldsymbol{\rho}}_i, 0)$, normalised by its own total with the log-depth scalar left raw, as the counts-level answer to the amortisation gap; an attention sink, a fixed null logit joining every destination’s softmax so that the aggregate saturates in the number of neighbours instead of ignoring it; coupled weight decay in the optimiser	each is available behind a flag and off in every fit reported here. The encoder correction is retained as an ablation rather than as the model, since it changes the evidence $\mathbf{z}$ is formed from; the sink cannot be exercised on a Delaunay graph, whose degree is nearly constant, and is kept for graphs whose degree varies; weight decay is recorded because it moves the gauge of Section 2.8 rather than removing it
Not built	Dirichlet-multinomial likelihood; per-cell counterfactuals with abduction; bounded conditional $\sigma_w(\mathbf{c}, t)$; estimation of κ from a nuclear/extranuclear split; edge features in the attention layer	each has a stated trigger that has not fired, or is future work. The <i>type-level</i> counterfactual, in which the context prior is moved between two neighbourhoods with the intrinsic state held fixed, is built and exercised; only the per-cell form, which needs a non-degenerate response posterior, is not